

Multidimensional Goodhart

Response Channels, Scorecards, and Residual Shape

Kerkko ‘Xylix’ Pelttari

Abstract. The initial hypothesis was that multidimensional proxy optimization would push residual error into more structured, perhaps more complex, hidden directions. The current framework narrows that into a conditional claim: residual shape is governed by the response channel, action/search geometry, aggregation rule, and declared hidden value model. Selection over fixed states licenses baseline response curves and value-drift bounds. Intervention at fixed type requires action, cost, search, and stakes primitives. Multidimensional scorecards have exchange-rate and entry effects, not a generic “more metrics” sign. The transfer rule is a response-modeling contract rather than a universal Goodhart theorem.

Motivation

Scalar Goodhart warnings compress two things: the target is reduced to a proxy, and the residual error is often reduced to an unnamed loss. In multidimensional settings the residual is the object of interest. It has direction, support, active constraints, exchange rates, and evidence requirements. The question is therefore not whether optimization can break proxies, but which model predicts where the error vector moves.

Use $G(s) \in \mathbb{R}^m$ for target-relevant state, $P(s) \in \mathbb{R}^k$ for proxy features, φ for the intended proxy relation, and $\varepsilon(s) = P(s) - \varphi(G(s))$ for proxy residual. A claim about multidimensional Goodhart must say how optimization pressure changes the law of s or the behavior of fixed types.

Current Findings

- **Selection response.** A pure selection channel reweights a fixed baseline law: $\mu_\theta = L_\theta \mu$. Hidden drift is $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu[H]}$. Threshold response and Boltzmann response are baseline path functionals; covariance is only a local velocity where the tilted moment-generating function is finite.
- **Selection/value bounds.** If $\delta = \|L_\theta - 1\|_{L^2(\mu)}$ and hidden coordinates have standard deviations s_i , then $|B_{H_i}| \leq \delta s_i$ and $\|B_H\|_2 \leq \delta \|s\|_2$ after declaring the coordinate norm. For a declared scalar value vector v , $|\Delta(v \cdot H)| \leq \delta \sqrt{v^T \Sigma_H v}$. The value metric is a modeling input, not inferred from proxy movement.
- **Intervention channels.** Intervention means the response kernel changes at fixed type: $K_{\theta(ds|u)} \neq K_0(ds|u)$ on a positive-mass set. The one-dimensional quadratic threshold toy gives a gaming wedge $\Delta = \sqrt{2\kappa V}$; this is a signature of ease of gaming and stakes, not a baseline covariance result. For convex action cost c and score deficit d , the private score-deficit cost is $m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)]$. This is not a welfare bound without a hidden-harm functional.
- **Scorecards and exchange rates.** With additive score, separable quadratic costs, fixed deficit d , score weights w_j , and hidden harm rates h_j , fixed-deficit per-agent harm is $H_{\text{per}(M,d)} = d \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}$. Conservation across active measured sets requires $h_j = c w_j$. Population harm is a separate entry object; conjunctive aggregation changes the comparison again.
- **Response shape.** Proxy pressure does not generically select minimum-complexity residuals. Quadratic costs select cost-minimal directions; fixed charges and caps can create lumpy active-set switches; low-rank affordances constrain drift to an image; search-prior claims require a predeclared coding or search process.

Why the Framework Matters

The framework replaces slogan transfer with primitive transfer. A domain application must declare type space U , baseline behavior, policy exposure, selection weights W_θ and/or response kernels K_θ , action/search geometry, proxy/target relation, aggregation, hidden harm/value, and evidence standard. The same score movement can be selection, harmful fixed-type gaming, harmless proxy repair, real improvement, or a mixture. Without the contract, a Goodhart explanation is often just a retrospective label.

This matters most when importing formal analogues. Lucas-style policy non-invariance, strategic classification, multitask incentives, adaptive holdout reuse, reward-model overoptimization, and benchmark contamination are nearby warnings, but each supplies only some primitives. None by itself gives a hidden welfare model, an action-cost geometry, an empirical estimate of κ , or a scorecard exchange rate. The contract says which calculation is licensed: a selection response curve, a reweighting bound, an intervention budget, an aggregation comparison, or a shape prediction.

The framework is also intentionally lossy about policy. It can say that a toy diagnostic is unavailable when effective weights, hidden harm, response ease, or stakes are not declared. It can say that aggregate score movement is insufficient evidence for hidden-target improvement. It cannot rank policies or infer welfare from score movement without additional empirical and value commitments.

Live Boundaries and Next Work

- The deterministic adaptive-hardening boundary is closed only in the finite-channel additive-score, separable-quadratic, fixed- M , fixed- d , fixed- V , fixed-weight, deterministic-observation contract: fixed-deficit gaming stops exactly when $S_{t(M)} = \sum_j \kappa_{j,t} w_j^2 < \frac{d^2}{2V}$.
- Stochastic observation, repair-budget design, endogenous stakes, changing measured sets or deficits, shared bottlenecks, nonconvex geometry, and policy optimization remain separate dynamic problems.
- Empirical applications require repeated-type observations, exposure variation, action traces, cost variation, or predeclared hidden outcomes. The framework does not license hidden-target estimates, welfare or research-value inference, RLHF claims, or generic advice that more metrics are good or bad without those primitives.